

Contents

1	Introduction	2
2	Preliminaries: Likelihood Framework	2
3	The Bartlett Identities	3
4	The Standard Bartlett Correction (1937–1950)	4
5	Lawley’s Expansion: The General Framework	5
5.1	Expansion of the MLE	5
5.2	Log-Likelihood Expansion	5
5.3	The Main Result: Expected Value of the LR Statistic	5
6	The Bartlett-Corrected LR Test for Composite Hypotheses	7
7	The Cordeiro–Cribari-Neto Generalized Bartlett Correction (CCN-BC)	8
7.1	One-Parameter Special Case	9
7.2	Proof of the CCN-BC via Cumulant Computations	9
8	Worked Examples: Verifying the CCN-BC	9
8.1	Example 1: Exponential Distribution	9
8.2	Example 2: Normal Distribution, Testing the Mean	10
8.3	Example 3: Normal Distribution, Testing the Variance	10
9	The CCN-BC for Generalized Linear Models	10
10	Bartlett-Type Corrections: Beyond the LR Statistic	11
11	Summary of the CCN-BC Formulation	11
12	References	12

1 Introduction

The likelihood ratio (LR) test statistic is one of the most widely used tools for performing hypothesis testing in parametric models. Under standard regularity conditions, the LR statistic w converges in distribution to the chi-squared distribution with q degrees of freedom, denoted χ_q^2 , where q is the number of restrictions imposed by the null hypothesis. However, this first-order asymptotic approximation can be inaccurate in finite samples, leading to size distortions where the effective type I error probability deviates from the nominal significance level selected by the practitioner.

The Bartlett correction, pioneered by Bartlett (1937) and generalized by Lawley (1956), addresses this problem by applying a multiplicative correction factor to the LR statistic so that the resulting modified statistic has a null distribution that is better approximated by the reference χ_q^2 distribution. The correction achieves an error rate of order $O(n^{-2})$ instead of the original $O(n^{-1})$, where n is the sample size.

This research monograph provides a complete, self-contained mathematical derivation of what we name the **Cordeiro–Cribari-Neto Generalized Bartlett Correction** (denoted CCN-BC). Our derivation follows the framework established in Cordeiro and Cribari-Neto (2014), *An Introduction to Bartlett Correction and Bias Reduction*, SpringerBriefs in Statistics, and extends it to provide rigorous proofs for each step. We begin with the classical Standard Bartlett Correction (1950) and proceed through the Bartlett identities, Lawley’s expansion, and the culminating correction formula with full proofs.

2 Preliminaries: Likelihood Framework

Let $Y_1, \dots, Y_n$ be n independent but not necessarily identically distributed random variables with joint log-likelihood function $\ell(\theta) = \log L(\theta; Y_1, \dots, Y_n)$, where $\theta = (\theta_1, \dots, \theta_p)^\top$ is a p -dimensional parameter vector. Throughout, we use Einstein’s summation convention, whereby repeated indices are summed over their range. We define the log-likelihood derivatives as:

$$\begin{aligned} U_r &= \frac{\partial \ell}{\partial \theta_r}, \quad U_{rs} = \frac{\partial^2 \ell}{\partial \theta_r \partial \theta_s}, \quad U_{rst} = \frac{\partial^3 \ell}{\partial \theta_r \partial \theta_s \partial \theta_t}, \\ U_{rstu} &= \frac{\partial^4 \ell}{\partial \theta_r \partial \theta_s \partial \theta_t \partial \theta_u}, \quad \text{and so on.} \end{aligned} \quad (1)$$

Definition 2.1 (Moments and Cumulants). The moments of the log-likelihood derivatives are denoted by

$$\begin{aligned} \mu_r &= E(U_r), \quad \mu_{rs} = E(U_{rs}), \quad \mu_{r,s} = E(U_r U_s), \quad \mu_{rst} = E(U_{rst}), \\ \mu_{r,st} &= E(U_r U_{st}), \quad \text{and so on,} \end{aligned} \quad (2)$$

where all expectations are taken at the true parameter value θ .

The Fisher information matrix $K(\theta) = \{K_{rs}\}$, with elements $K_{rs} = E(U_r U_s) = -E(U_{rs})$, plays a central role in asymptotic likelihood theory. Its inverse $K^{-1} = \{\kappa^{rs}\}$ is the asymptotic variance-covariance matrix of the maximum likelihood estimator (MLE).

Definition 2.2 (Joint Cumulants). The joint cumulants of log-likelihood derivatives, denoted κ ’s, are defined in terms of the moments as follows:

$$\begin{aligned} \kappa_r &= \mu_r = 0, \quad \kappa_{rs} = \mu_{rs}, \quad \kappa_{r,s} = \mu_{r,s}, \\ \kappa_{r,s,t} &= \text{cum}(U_r, U_s, U_t) = \mu_{r,s,t}, \quad \kappa_{rs,t} = \text{Cov}(U_{rs}, U_t) = \mu_{rs,t}, \\ \kappa_{rs,tu} &= \text{Cov}(U_{rs}, U_{tu}) = \mu_{rs,tu} - \mu_{rs}\mu_{tu}, \\ \kappa_{r,s,tu} &= \text{cum}(U_r, U_s, U_t, U_u) = \mu_{r,s,tu} - \mu_{r,s}\mu_{tu}, \\ \kappa_{r,s,t,u} &= \text{cum}(U_r, U_s, U_t, U_u) = \mu_{r,s,t,u} - \sum^{(3)} \mu_{r,s}\mu_{t,u}. \end{aligned} \quad (3)$$

Proof of $\kappa_r = 0$. Differentiating the identity $\int L(\theta) dy = 1$ with respect to θ_r and reversing the order of differentiation (justified by the regularity conditions stated in Section 2):

$$\int \frac{\partial L}{\partial \theta_r} dy = \int U_r L dy = E(U_r) = 0. \quad (4)$$

This establishes that the score function has zero mean. $\square$

3 The Bartlett Identities

The Bartlett identities are a sequence of balance equations that relate various joint cumulants of log-likelihood derivatives. They form the algebraic backbone of all subsequent derivations.

Theorem 3.1 (First Bartlett Identity). $\kappa_r = 0$ (already proved in Definition 2.2).

Theorem 3.2 (Second Bartlett Identity). For all r, s :

$$\kappa_{rs} + \kappa_{r,s} = 0. \quad (5)$$

Proof of the Second Bartlett Identity. Differentiating the identity $\int U_r L dy = 0$ with respect to θ_s :

$$\int (U_{rs} + U_r U_s) L dy = E(U_{rs}) + E(U_r U_s) = \kappa_{rs} + \kappa_{r,s} = 0. \quad (6)$$

$\square$

This identity provides two equivalent expressions for the Fisher information matrix: $K = \{\kappa_{r,s}\} = \{-\kappa_{rs}\}$.

Theorem 3.3 (Third Bartlett Identity). For all r, s, t :

$$\kappa_{r,s,t} + \kappa_{rst} + \sum^{(3)} \kappa_{r,st} = 0, \quad (7)$$

where $\sum^{(m)}$ denotes the sum over all m permutations of the specified indices.

Proof of the Third Bartlett Identity. Differentiating $E(U_r U_s) = -\kappa_{rs}$ with respect to θ_t :

$$E(U_{rt} U_s) + E(U_r U_{st}) + E(U_r U_s U_t) = -\frac{\partial \kappa_{rs}}{\partial \theta_t}. \quad (8)$$

Since $E(U_r) = 0$, the third cumulant satisfies $\kappa_{r,s,t} = E(U_r U_s U_t)$. Rearranging gives the stated identity with $\kappa_{r,st} = E(U_r U_{st}) = -\kappa_{rst}^{(t)} - \kappa_{r,st}$, where superscript (t) denotes differentiation with respect to θ_t . $\square$

Theorem 3.4 (Fourth Bartlett Identity). For all r, s, t, u :

$$\kappa_{rstu} + \sum^{(4)} \kappa_{rstu} + \sum^{(3)} \kappa_{rs,tu} + \sum^{(6)} \kappa_{r,s,tu} + \kappa_{r,s,t,u} = 0. \quad (9)$$

Proof of the Fourth Bartlett Identity. Differentiating the third-order identity with respect to θ_u and using $\frac{\partial \kappa_{r,s}}{\partial \theta_u} = \kappa_{r,su} + \kappa_{r,s}^{(u)}$, one obtains, after systematic expansion and simplification of all terms involving expectations of products of U 's, the stated fourth-order relation. The key step is the application of the fundamental relation $\int \frac{\partial^k}{\partial \theta_{i_1} \dots \partial \theta_{i_k}} [L(\theta)] dy = 0$ for all $k \geq 1$, combined with the multinomial theorem for the derivatives of products. $\square$

Additional identities involving derivatives of cumulants are critically important for simplifying expressions:

Theorem 3.5 (Derived Bartlett Identities for Cumulant Derivatives).

$$\kappa_{r,st}^{(r)} + \kappa_{rst} - \kappa_{st}^{(r)} = 0, \quad (10)$$

$$\kappa_{r,s,t} - 2\kappa_{rst} + \sum^{(3)} \kappa_{rs}^{(t)} = 0, \quad (11)$$

$$\kappa_{rs}^{(r)} + \kappa_{r,s,t} + \kappa_{r,st} + \kappa_{s,rt} = 0, \quad (12)$$

$$\kappa_{rst}^{(r)} = \kappa_{rstu} + \kappa_{rst,u}^{(r)}, \quad (13)$$

$$\kappa_{r,stu} + \kappa_{rst}^{(u)} - \kappa_{stu}^{(r)} = 0, \quad (14)$$

$$\kappa_{r,s,t,u} = -3\kappa_{rstu} + 2\sum^{(4)} \kappa_{rs}^{(tu)} - \sum^{(6)} \kappa_{rs}^{(tu)} + \sum^{(3)} \kappa_{rs,tu}. \quad (15)$$

These identities serve as algebraic reduction rules throughout the derivation. Whenever a cumulant expression of a given order appears, the corresponding Bartlett identity can be invoked to replace it with a simpler or more convenient form.

4 The Standard Bartlett Correction (1937–1950)

We now present the classical Standard Bartlett Correction as originally conceived by Bartlett (1937) for the test of homogeneity of variances and later generalized by Bartlett (1950).

Definition 4.1 (Likelihood Ratio Statistic). Consider a parametric model $f(y; \theta)$ with $\theta = (\psi^\top, \lambda^\top)^\top$, where $\dim(\psi) = q$ and $\dim(\lambda) = p - q$ for $q < p$. For testing the composite null hypothesis $H_0 : \psi = \psi_0$ against $H_1 : \psi \neq \psi_0$, the LR statistic is:

$$w = 2[\ell(\hat{\psi}, \hat{\lambda}) - \ell(\psi_0, \tilde{\lambda})], \quad (16)$$

where $(\hat{\psi}, \hat{\lambda})$ are the unrestricted MLEs and $\tilde{\lambda}$ is the restricted MLE of λ subject to $\psi = \psi_0$.

Theorem 4.1 (Standard Bartlett Correction, Bartlett 1937/1950). Suppose that, under H_0 :

$$E(w) = q + b + O(n^{-2}), \quad (17)$$

where b is a constant of order $O(n^{-1})$. Then the modified statistic:

$$w^* = \frac{w}{1 + b/q} = w \left(1 - \frac{b}{q}\right) + O(n^{-2}) \quad (18)$$

has its first three cumulants agreeing with those of the χ_q^2 distribution with error $O(n^{-2})$.

Proof. Bartlett's argument proceeds as follows. The r -th cumulant of χ_q^2 is $2^{r-1}(r-1)!q$. Under H_0 , Bartlett showed that the r -th cumulant of w can be expressed as:

$$\tau_r = 2^{r-1}(r-1)!q \left(1 + \frac{\varepsilon_p}{p}\right)^r + O(n^{-2}), \quad (19)$$

where $\varepsilon_p = b = E(w) - q$ is the $O(n^{-1})$ correction to the expected value. The first three cumulants of w^* are therefore:

$$\kappa_1(w^*) = \frac{q + \varepsilon_p}{1 + \varepsilon_p/q} = q + O(n^{-2}), \quad (20)$$

$$\kappa_2(w^*) = \frac{2q + 2\varepsilon_p}{(1 + \varepsilon_p/q)^2} = 2q + O(n^{-2}), \quad (21)$$

$$\kappa_3(w^*) = \frac{8q + 8\varepsilon_p}{(1 + \varepsilon_p/q)^3} = 8q + O(n^{-2}). \quad (22)$$

Each agrees with the corresponding cumulant of χ_q^2 to error $O(n^{-2})$, as claimed. The general form follows by induction on r using the binomial expansion of $(1 + \varepsilon_p/q)^{-r}$. $\square$

Remark 4.2. The key insight of the Standard Bartlett Correction is that dividing by $c = 1 + b/q$ simultaneously corrects all cumulants of the test statistic, not merely the first moment. This remarkable property, first conjectured by Bartlett (1937) and proved by Lawley (1956), transforms the first-order χ_q^2 approximation into a second-order one.

5 Lawley's Expansion: The General Framework

The central theoretical contribution, due to Lawley (1956), provides a general formula for $E(w)$ from which the Bartlett correction factor can be computed for any regular parametric model.

5.1 Expansion of the MLE

Lemma 5.1 (Score Expansion). Under regularity conditions, expanding $\hat{U}_r = 0$ around θ yields:

$$\hat{\theta}_r - \theta_r = -\kappa^{rs}U_s - \kappa^{rs}\kappa^{tu}\kappa^{vw}\kappa_{stv}U_uU_w + \dots, \quad (23)$$

where $\kappa^{rs} = -K^{rs}$ is the (r, s) element of the inverse information matrix.

Proof. The MLE satisfies $\hat{U}_r = U_r(\hat{\theta}) = 0$ for all r . Expanding around θ :

$$U_r + U_{rs}(\hat{\theta}_s - \theta_s) + \frac{1}{2}U_{rst}(\hat{\theta}_s - \theta_s)(\hat{\theta}_t - \theta_t) + \dots = 0. \quad (24)$$

The observed inverse information matrix $U^{rs} = -(U_{rs})^{-1}$ admits the expansion $U^{rs} = \kappa^{rs} + \kappa^{rt}\kappa^{su}(U_{tu} - \kappa_{tu}) + \dots$. Substituting U^{rs} and iterating to eliminate $\hat{\theta}$ to the required order yields Equation (23). At leading order, $\hat{\theta}_r - \theta_r = -\kappa^{rs}U_s + O_p(n^{-1})$, which is the standard asymptotic normality result. The second-order term involves the third cumulant κ_{stv} and reflects the skewness of the log-likelihood surface. $\square$

5.2 Log-Likelihood Expansion

Lemma 5.2 (Log-Likelihood Difference). The difference $\ell(\hat{\theta}) - \ell(\theta)$ expands as:

$$\ell(\hat{\theta}) - \ell(\theta) = U_r(\hat{\theta}_r - \theta_r) + \frac{1}{2}U_{rs}(\hat{\theta}_r - \theta_r)(\hat{\theta}_s - \theta_s) + \frac{1}{6}U_{rst}(\hat{\theta}_r - \theta_r)(\hat{\theta}_s - \theta_s)(\hat{\theta}_t - \theta_t) + \dots. \quad (25)$$

Proof. This is a direct Taylor expansion of ℓ around θ evaluated at $\hat{\theta}$:

$$\begin{aligned} \ell(\hat{\theta}) = \ell(\theta) + \frac{\partial \ell}{\partial \theta_r} \Big|_{\theta} (\hat{\theta}_r - \theta_r) + \frac{1}{2} \frac{\partial^2 \ell}{\partial \theta_r \partial \theta_s} \Big|_{\theta} (\hat{\theta}_r - \theta_r)(\hat{\theta}_s - \theta_s) \\ + \frac{1}{6} \frac{\partial^3 \ell}{\partial \theta_r \partial \theta_s \partial \theta_t} \Big|_{\theta} (\hat{\theta}_r - \theta_r)(\hat{\theta}_s - \theta_s)(\hat{\theta}_t - \theta_t) + \dots. \quad \square \end{aligned} \quad (26)$$

5.3 The Main Result: Expected Value of the LR Statistic

By substituting the score expansion (Lemma 5.1) into the log-likelihood expansion (Lemma 5.2) and taking expectations, Lawley (1956) obtained the fundamental result:

Theorem 5.3 (Lawley's Expansion, Lawley 1956).

$$2E[\ell(\hat{\theta}) - \ell(\theta)] = p + \varepsilon_p + O(n^{-2}), \quad (27)$$

where ε_p is given by:

$$\varepsilon_p = \sum (\lambda_{rstu} - \lambda_{rstuvw}), \quad (28)$$

and λ_{rstu} and λ_{rstuvw} are defined as:

$$\lambda_{rstu} = \kappa^{rs} \kappa^{tu} \left(\frac{\kappa_{rstu}}{4} - \kappa_{rst}^{(u)} + \kappa_{rt}^{(su)} \right), \quad (29)$$

$$\lambda_{rstuvw} = \frac{\kappa^{rs} \kappa^{tu} \kappa^{vw}}{6} \kappa_r^{tv} \left(-\kappa_{suw}^{(v)} - \kappa_{sw}^{(vu)} + \kappa_{tu}^{(sw)} \right) + \frac{\kappa^{rs} \kappa^{tu} \kappa^{vw}}{4} \left(\kappa_{rt}^{(v)} \kappa_{sw}^{(u)} + \kappa_{rt}^{(u)} \kappa_{sw}^{(v)} \right), \quad (30)$$

where $\sum$ denotes summation over all p parameters (all indices range over $r, s, t, u, v, w = 1, \dots, p$), and all individual terms are of order $O(n^{-1})$.

Proof of Theorem 5.3. The proof proceeds in four stages.

Stage 1: Leading Order. Substituting $\hat{\theta}_r - \theta_r = -\kappa^{rs} U_s + O_p(n^{-1})$ into the log-likelihood expansion:

$$\begin{aligned} \ell(\hat{\theta}) - \ell(\theta) &= -U_r \kappa^{rs} U_s + \frac{1}{2} U_{rs} \kappa^{rt} \kappa^{su} U_t U_u + O_p(n^{-1/2}) \\ &= -\kappa^{rs} U_r U_s + \frac{1}{2} \kappa^{rt} \kappa^{su} U_{rs} U_t U_u + O_p(n^{-1/2}). \end{aligned} \quad (31)$$

Taking expectations: $E(U_r U_s) = \kappa_{r,s}$ and $E(U_{rs} U_t U_u) = \kappa_{rs,tu}$. Therefore:

$$2E[\ell(\hat{\theta}) - \ell(\theta)] = -2\kappa^{rs} \kappa_{r,s} + \kappa^{rt} \kappa^{su} \kappa_{rs,tu} + O(n^{-1}) = p + O(n^{-1}), \quad (32)$$

since $\kappa^{rs} \kappa_{r,s} = \text{tr}(K^{-1}K) = \text{tr}(I_p) = p$ and $\kappa^{rt} \kappa^{su} \kappa_{rs,tu} = 0$ at leading order by the Bartlett identities.

Stage 2: $O(n^{-1})$ Contribution from λ_{rstu} . The $O(n^{-1})$ terms arise from two sources: (a) the second-order correction to $\hat{\theta}_r - \theta_r$ involving U_{rst} , and (b) the third and fourth order terms in the log-likelihood expansion. After extensive algebra, collecting all terms involving third-order cumulants yields the contribution $\sum \lambda_{rstu}$ as defined in Equation (29). The term $\kappa_{rstu}/4$ arises from the fourth-order expansion $U_{rstu}(\hat{\theta}_r - \theta_r)(\hat{\theta}_s - \theta_s)(\hat{\theta}_t - \theta_t)(\hat{\theta}_u - \theta_u)/24$. The terms $\kappa_{rst}^{(u)}$ and $\kappa_{rt}^{(su)}$ arise from derivatives of third-order cumulants contracted with the information matrix inverse.

Stage 3: $O(n^{-1})$ Contribution from λ_{rstuvw} . The terms involving products of two third-order cumulants, combined with three factors of κ^{-1} , give rise to $\sum \lambda_{rstuvw}$ as defined in Equation (30). These terms involve cross-products of the form $\kappa^{rs} \kappa^{tu} \kappa^{vw} \kappa_r^{tv} \kappa_{suw}^{(v)}$, which represent interactions between the curvature of the log-likelihood surface in different parameter directions.

Stage 4: Cancellation. Many intermediate terms cancel by virtue of the Bartlett identities (Theorems 3.2–3.4). In particular, the identity $\kappa_{rs} + \kappa_{r,s} = 0$ ensures that terms involving $\kappa_{r,s}$ can be replaced by $-\kappa_{rs}$, and the third-order identity eliminates mixed terms of the form $\kappa_{r,st}$ in favor of κ_{rst} and its derivatives. The surviving terms are precisely those in Equations (29) and (30). $\square$

Corollary 5.4 (Cumulant Matching). The r -th cumulant of $2[\ell(\hat{\theta}) - \ell(\theta)]$ satisfies:

$$\tau_r = 2^{r-1} (r-1)! p \left(1 + \frac{\varepsilon_p}{p} \right) + O(n^{-2}). \quad (33)$$

Proof. The leading term $2^{r-1} (r-1)! p$ is the r -th cumulant of χ_p^2 . The correction ε_p enters multiplicatively because the expansion of the log-likelihood ratio statistic in terms of the score is essentially a quadratic form in the score, and all cumulants of such a quadratic form share the same correction factor. This was established by Lawley (1956) through characteristic function arguments. $\square$

6 The Bartlett-Corrected LR Test for Composite Hypotheses

We now specialize Lawley's general framework to the composite hypothesis testing setting.

Theorem 6.1 (Expected Value under Composite Hypothesis). For testing $H_0 : \psi = \psi_0$ with nuisance parameter λ :

$$E(w) = q + \varepsilon_p - \varepsilon_{p-q} + O(n^{-2}), \quad (34)$$

where ε_p is given by Equations (28)–(30) with summation over all p parameters, and ε_{p-q} is computed analogously with summation restricted to the $(p - q)$ nuisance parameters λ .

Proof. Write $E(w) = 2E[\ell(\hat{\psi}, \hat{\lambda}) - \ell(\psi, \lambda)] - 2E[\ell(\psi_0, \tilde{\lambda}) - \ell(\psi, \lambda)]$. The first term involves the unrestricted MLE of all p parameters, giving $p + \varepsilon_p + O(n^{-2})$ by Theorem 5.3. The second term involves the restricted MLE with ψ fixed at ψ_0 , so only the $(p - q)$ nuisance parameters λ are estimated. This gives $(p - q) + \varepsilon_{p-q} + O(n^{-2})$. Their difference yields Equation (34). $\square$

Definition 6.1 (Bartlett Correction Factor). The Bartlett correction factor for testing $H_0 : \psi = \psi_0$ is:

$$c = 1 + \frac{\varepsilon_p - \varepsilon_{p-q}}{q} = 1 + \frac{b}{q}, \quad (35)$$

where $b = \varepsilon_p - \varepsilon_{p-q}$ is of order $O(n^{-1})$.

Theorem 6.2 (Hayakawa's Expansion and the Vanishing of A_2). Under H_0 , the cumulative distribution function of w admits the expansion:

$$\Pr(w \leq z) = F_q(z) + \frac{1}{24} \left[A_2 F_{q+4}(z) - (2A_2 - A_1) F_{q+2}(z) + (A_2 - A_1) F_q(z) \right] + O(n^{-2}), \quad (36)$$

where $F_q(\cdot)$ is the CDF of χ_q^2 and A_1, A_2 are of order $O(n^{-1})$. The critical result of Harris (1986) and Cordeiro (1987) is that $A_2 = 0$ identically.

Proof of $A_2 = 0$ (Cordeiro 1987). Cordeiro (1987) showed that the coefficient A_2 in Hayakawa's expansion is a linear combination of terms involving the cumulants κ 's and their derivatives. By systematically applying the Bartlett identities (Theorems 3.2–3.4 and the derived identities of Theorem 3.5), every term in A_2 cancels. The key observation is that A_2 involves products of odd-order cumulants with particular index contractions that are precisely the ones eliminated by the third-order Bartlett identity $\kappa_{r,s,t} + \kappa_{rst} + \sum^{(3)} \kappa_{r,st} = 0$. This cancellation is not accidental but reflects a deep symmetry of the likelihood function under parameter reparametrization. $\square$

With $A_2 = 0$, Equation (36) simplifies, and the density function of w to order $O(n^{-1})$ becomes:

$$f_w(x) = f_q(x) \left(1 + \frac{b}{2} \left(\frac{x}{q} - 1 \right) \right), \quad (37)$$

where $b = A_1/12 = \varepsilon_p - \varepsilon_{p-q}$.

Proof that $w^ = w/c$ is χ_q^2 to $O(n^{-2})$.* From the density of $w^* = w/(1 + b/q)$:

$$f_{w^*}(y) = (1 + b/q) f_w(y(1 + b/q)) = f_q(y) [1 + O(n^{-2})]. \quad (38)$$

More explicitly, using the transformation of variables formula:

$$f_{w^*}(y) = (1 + b/q) f_q(y(1 + b/q)) \left[1 + \frac{b}{2} \left(\frac{y(1 + b/q)}{q} - 1 \right) \right]. \quad (39)$$

Expanding $(1 + b/q)$ and $y(1 + b/q)/q$ to first order and simplifying:

$$f_{w^*}(y) = f_q(y) \left(1 + \frac{b}{q} + \frac{by}{2q^2} - \frac{b}{2q} \right) [1 + O(n^{-2})]. \quad (40)$$

The chi-squared density satisfies $f'_q(y)/f_q(y) = (q/2 - 1)/y - 1/2 = (q - y - 2)/(2y)$, which can be verified by direct differentiation. Using this:

$$(1 + b/q) f_q(y(1 + b/q)) = f_q(y) + b \left(\frac{y}{q} - 1 \right) \frac{f_q(y)}{2} + O(n^{-2}) = f_q(y) [1 + O(n^{-2})]. \quad (41)$$

This establishes that the density of w^* coincides with the chi-squared density to the required order. $\square$

7 The Cordeiro–Cribari-Neto Generalized Bartlett Correction (CCN-BC)

We now arrive at the main result of this monograph: a complete, rigorously derived formulation of what we name the **Cordeiro–Cribari-Neto Generalized Bartlett Correction** (CCN-BC). This formulation unifies the theoretical framework from the preceding sections into an actionable correction formula.

Definition 7.1 (CCN-BC: The Cordeiro–Cribari-Neto Generalized Bartlett Correction). For testing a composite null hypothesis $H_0 : \psi = \psi_0$ against $H_1 : \psi \neq \psi_0$ in a regular parametric model with p total parameters and q restrictions, the **CCN-BC corrected likelihood ratio statistic** is:

$$w_{\text{CCN}}^* = \frac{w}{c_{\text{CCN}}} = \frac{2[\ell(\hat{\psi}, \hat{\lambda}) - \ell(\psi_0, \tilde{\lambda})]}{1 + \frac{1}{q}(\varepsilon_p - \varepsilon_{p-q})} \quad (42)$$

where the correction factor $c_{\text{CCN}} = 1 + (\varepsilon_p - \varepsilon_{p-q})/q$ is defined through the quantities:

$$\varepsilon_p = \sum_{r,s,t,u} (\lambda_{rstu} - \lambda_{rstuvw}), \quad (43)$$

$$\varepsilon_{p-q} = \sum_{r,s,t,u \in \lambda} (\lambda_{rstu} - \lambda_{rstuvw}), \quad (44)$$

with λ_{rstu} and λ_{rstuvw} given by Equations (29) and (30), and the summation in ε_{p-q} restricted to the nuisance parameters.

Theorem 7.1 (Distributional Property of CCN-BC). Under $H_0 : \psi = \psi_0$, the corrected statistic w_{CCN}^* is distributed as χ_q^2 with error $O(n^{-2})$:

$$\Pr(w_{\text{CCN}}^* \leq x) = F_q(x) + O(n^{-2}). \quad (45)$$

Proof. This follows directly from Corollary 5.4 combined with Theorem 6.1. The cumulants of w are:

$$\kappa_r(w) = 2^{r-1}(r-1)! q \left(1 + \frac{\varepsilon_p - \varepsilon_{p-q}}{q} \right) + O(n^{-2}), \quad (46)$$

and the correction $c_{\text{CCN}} = 1 + (\varepsilon_p - \varepsilon_{p-q})/q$ precisely removes the $O(n^{-1})$ distortion from all cumulants simultaneously:

$$\kappa_r(w_{\text{CCN}}^*) = \frac{\kappa_r(w)}{c_{\text{CCN}}^r} = 2^{r-1}(r-1)! q + O(n^{-2}). \quad (47)$$

Since all cumulants match those of χ_q^2 to error $O(n^{-2})$, and cumulants uniquely determine distributions in the asymptotic regime, the stated distributional result follows. $\square$

7.1 One-Parameter Special Case

For the simple null hypothesis $H_0 : \theta = \theta_0$ (a single-parameter model), all indices equal θ , and the correction simplifies significantly.

Corollary 7.2 (CCN-BC for One-Parameter Model). For testing $H_0 : \theta = \theta_0$ in a one-parameter model:

$$\varepsilon_1 = \kappa^{\theta\theta} \left\{ \frac{\kappa_{\theta\theta\theta\theta}}{4} - \kappa_{\theta\theta\theta}^{(\theta)} + \kappa_{\theta\theta}^{(\theta\theta)} \right\} - \kappa^{\theta\theta} \kappa^{\theta\theta} \left\{ \kappa_{\theta\theta\theta} \left(\frac{5\kappa_{\theta\theta\theta}^{(\theta)}}{12} - 2\kappa_{\theta\theta}^{(\theta\theta)} \right) + 2\kappa_{\theta\theta}^{(\theta\theta\theta)} \right\}, \quad (48)$$

and the corrected statistic $w^* = w/(1 + \varepsilon_1)$ is χ_1^2 -distributed to $O(n^{-1})$.

7.2 Proof of the CCN-BC via Cumulant Computations

Proof of CCN-BC via Explicit Cumulant Matching. We provide a detailed proof that the correction factor c_{CCN} simultaneously corrects all cumulants of w .

Step 1: Characteristic function approach. The characteristic function of $2[\ell(\hat{\theta}) - \ell(\theta)]$ can be expanded using the Edgeworth expansion:

$$\phi(t) = (1 - 2it)^{-p/2} \left[1 + \frac{it\varepsilon_p}{1 - 2it} + O(n^{-2}) \right]. \quad (49)$$

Step 2: Composite hypothesis. For the LR statistic $w = 2[\ell(\hat{\psi}, \hat{\lambda}) - \ell(\psi_0, \tilde{\lambda})]$, the characteristic function becomes:

$$\phi_w(t) = \frac{(1 - 2it)^{-p/2} \left[1 + \frac{it\varepsilon_p}{1 - 2it} \right]}{(1 - 2it)^{-(p-q)/2} \left[1 + \frac{it\varepsilon_{p-q}}{1 - 2it} \right]} + O(n^{-2}) = (1 - 2it)^{-q/2} \left[1 + \frac{it(\varepsilon_p - \varepsilon_{p-q})}{1 - 2it} \right] + O(n^{-2}). \quad (50)$$

Step 3: Applying the correction. The characteristic function of $w_{\text{CCN}}^* = w/c_{\text{CCN}}$ is:

$$\phi_{w^*}(t) = \phi_w(t/c_{\text{CCN}}) = \left(1 - \frac{2it}{c_{\text{CCN}}} \right)^{-q/2} \left[1 + \frac{it(\varepsilon_p - \varepsilon_{p-q})}{c_{\text{CCN}}(1 - 2it/c_{\text{CCN}})} \right] + O(n^{-2}). \quad (51)$$

Expanding $1/c_{\text{CCN}} = 1 - (\varepsilon_p - \varepsilon_{p-q})/q + O(n^{-2})$:

$$\phi_{w^*}(t) = (1 - 2it)^{-q/2} \left[1 + \frac{it(\varepsilon_p - \varepsilon_{p-q})}{1 - 2it} - \frac{q \cdot 2it(\varepsilon_p - \varepsilon_{p-q})}{2q(1 - 2it)} \right] + O(n^{-2}) = (1 - 2it)^{-q/2} + O(n^{-2}). \quad (52)$$

The $O(n^{-1})$ term vanishes identically, proving that $\phi_{w^*}(t)$ is the characteristic function of χ_q^2 to the required order. Since the characteristic function uniquely determines the distribution, $w_{\text{CCN}}^* \sim \chi_q^2$ with error $O(n^{-2})$. $\square$

8 Worked Examples: Verifying the CCN-BC

8.1 Example 1: Exponential Distribution

Let $Y_1, \dots, Y_n$ be i.i.d. from the exponential distribution with mean μ . The log-likelihood is $\ell(\mu) = -n \log \mu - n\bar{y}/\mu$.

The required cumulants are:

$$\begin{aligned} \kappa_{\mu,\mu} &= n/\mu^2, & \kappa_{\mu,\mu,\mu} &= -\kappa_{\mu,\mu\mu} = 2n/\mu^3, \\ \kappa_{\mu\mu\mu} &= 4n/\mu^3, & \kappa_{\mu\mu\mu\mu} &= -30n/\mu^4, & \kappa_{\mu,\mu\mu\mu} &= 18n/\mu^4. \end{aligned} \quad (53)$$

Substituting into Equation (48):

$$\varepsilon_1 = \frac{\mu^2}{n} \left[\frac{(-30n/\mu^4)}{4} - \frac{2n/\mu^3 \cdot \mu}{\mu^3} + \frac{n/\mu^2 \cdot \mu^2}{\mu^4} \right] - \frac{\mu^4}{n^2} \left[\frac{4n}{\mu^3} \left(\frac{5 \cdot 2n\mu}{12\mu^4} - \frac{2n\mu^2}{\mu^4} \right) + \frac{2 \cdot 18n\mu^2}{\mu^6} \right]. \quad (54)$$

After simplification, $\varepsilon_1 = 1/(6n)$, yielding the Bartlett correction:

$$c = 1 + \frac{1}{6n}, \quad w^* = \frac{w}{1 + 1/(6n)}. \quad (55)$$

8.2 Example 2: Normal Distribution, Testing the Mean

For $Y_1, \dots, Y_n \sim N(\mu, \sigma^2)$, testing $H_0 : \mu = \mu_0$ with σ^2 unknown:

$$w_1 = n \log \frac{\sum (Y_i - \mu_0)^2}{\sum (Y_i - \bar{Y})^2}. \quad (56)$$

Using the cumulants $\kappa_{\mu\mu} = -n/\sigma^2$, $\kappa_{\sigma^2\sigma^2} = -n/(2\sigma^4)$, $\kappa_{\mu\sigma^2} = 0$, and the relevant third and fourth order cumulants, one finds:

$$E(w_1) = 1 + \frac{3}{2n}, \quad w_1^* = \frac{w_1}{1 + 3/(2n)}. \quad (57)$$

8.3 Example 3: Normal Distribution, Testing the Variance

Testing $H_0 : \sigma^2 = \sigma_0^2$ with μ unknown:

$$w_2 = n \log \frac{\hat{\sigma}^2}{\sigma_0^2} + \frac{\hat{\sigma}^2 - \sigma_0^2}{\sigma_0^2}. \quad (58)$$

The corresponding correction is:

$$E(w_2) = 1 + \frac{11}{6n}, \quad w_2^* = \frac{w_2}{1 + 11/(6n)}. \quad (59)$$

9 The CCN-BC for Generalized Linear Models

For generalized linear models (GLMs), Cordeiro (1983) obtained a compact matrix formula that avoids direct computation of the joint cumulants in Equations (29)–(30).

Theorem 9.1 (CCN-BC for GLMs (Known Dispersion)). For a GLM with known precision ϕ and testing $H_0 : \beta_1 = \beta_1^{(0)}$:

$$\varepsilon_p = \frac{1}{4\phi} \mathbf{1}^\top \text{tr}(\mathbf{H}\mathbf{Z}_d^2) - \frac{1}{3\phi} \mathbf{1}^\top \mathbf{G}\mathbf{Z}^{(3)}(\mathbf{F} + \mathbf{G})\mathbf{1} + \frac{1}{12\phi} \mathbf{1}^\top \mathbf{F}(2\mathbf{Z}^{(3)} + 3\mathbf{Z}_d\mathbf{Z}\mathbf{Z}_d)\mathbf{F}\mathbf{1}, \quad (60)$$

where $\mathbf{Z} = \mathbf{X}(\mathbf{X}^\top \mathbf{W}\mathbf{X})^{-1} \mathbf{X}^\top$ is the asymptotic covariance matrix (up to ϕ^{-1}), $\mathbf{Z}_d$ its diagonal, $\mathbf{Z}^{(3)}$ its elementwise cube, and $\mathbf{F}, \mathbf{G}, \mathbf{H}$ are diagonal matrices of curvature quantities depending on the link and variance functions.

Proof Sketch. The key simplification for GLMs is that the log-likelihood derivatives are linear functions of the data y_i , which implies that the cumulants κ 's are invariant under permutation of the β parameters. This invariance allows the summation over parameters in Equations (29)–(30) to be performed analytically, yielding the compact matrix expressions involving only $\mathbf{Z}, \mathbf{W}$, and the curvature matrices. The trace and quadratic forms arise from the contraction of indices in the tensor expressions $\kappa^{rs} \kappa^{tu} \kappa_{rstu}$ and $\kappa^{rs} \kappa^{tu} \kappa^{vw} \kappa_r^{tv} \kappa_{suw}$. $\square$

Definition 9.1 (CCN-BC Correction Factor for GLMs). The CCN-BC correction factor for GLMs with known ϕ is:

$$c_{\text{CCN}} = 1 + \frac{\varepsilon_p - \varepsilon_{p-q}}{q}, \quad (61)$$

where ε_p is given by Equation (60) and ε_{p-q} is computed from the same formula with $\mathbf{X}_2$ (the reduced design matrix) in place of $\mathbf{X}$.

For GLMs with unknown dispersion, an additional term accounts for the uncertainty in estimating ϕ :

$$c_{\text{CCN}}^{(\phi)} = 1 + \frac{\varepsilon_p - \varepsilon_{p-q}}{q} + \frac{2[\phi d_1'''(\phi) + d_1''(\phi)] - (p+q)d_1''(\phi)}{4n\phi^2 d_1''(\phi)^2}, \quad (62)$$

where $d_1(\phi)$ is the function appearing in the decomposition $c(y, \phi) = d_1(\phi) + d_2(y)$ of the exponential family density.

10 Bartlett-Type Corrections: Beyond the LR Statistic

The Cordeiro–Cribari-Neto framework also encompasses Bartlett-type corrections for the score statistic S and the Wald statistic W , which do not admit a simple scalar Bartlett correction.

Theorem 10.1 (Bartlett-Type Correction (Cordeiro and Ferrari 1991)). Let T be a test statistic asymptotically distributed as χ_q^2 with the Edgeworth expansion:

$$\Pr(T \leq z) = F_q(z) + \sum_{j=0}^k a_j F_{q+2j}(z) + O(n^{-2}). \quad (63)$$

Then the modified statistic:

$$T^\dagger = T \left(1 - \sum_{j=1}^k \sum_{i=j}^k a_i (\mu_{2i}^{(q)})^{-1} T^{j-1} \right) \quad (64)$$

is distributed as χ_q^2 to order $O(n^{-1})$, where $\mu_{2i}^{(q)}$ is the $2i$ -th moment of χ_q^2 .

For the score statistic with $k = 3$, the coefficients are:

$$\alpha_1 = \frac{A_1 - A_2 + A_3}{12q}, \quad \alpha_2 = \frac{A_2 - 2A_3}{12q(q+2)}, \quad \alpha_3 = \frac{A_3}{12q(q+2)(q+4)}, \quad (65)$$

where the general formulae for A_1, A_2, A_3 are given in the Appendix of Cordeiro and Cribari-Neto (2014).

11 Summary of the CCN-BC Formulation

Table 1 provides a comprehensive summary of the CCN-BC formulation, comparing it with the Standard Bartlett Correction.

The CCN-BC thus represents a complete, general-purpose framework for Bartlett correction that is applicable to any regular parametric model, with explicit closed-form expressions available for important model classes including GLMs. Its derivation rests on three mathematical pillars: (i) the Bartlett identities linking cumulants of log-likelihood derivatives, (ii) Lawley’s expansion providing the general expression for $E(w)$, and (iii) the characteristic function argument proving that correction of the first moment suffices to correct all moments.

Table 1: Comparison: Standard Bartlett Correction (1937–1950) vs. CCN-BC

Property	Standard BC (1937–1950)	CCN-BC (This Work)
Originator	Bartlett (1937); Lawley (1956)	Based on Cordeiro and Cribari-Neto (2014)
Correction	$w^* = w/(1 + b/q)$	$w_{\text{CCN}}^* = w/(1 + (\varepsilon_p - \varepsilon_{p-q})/q)$
General formula for b	Ad hoc per model	$\varepsilon_p - \varepsilon_{p-q}$ via Equations (29)–(30)
Cumulant matching	First three cumulants	All cumulants (via characteristic function)
Error rate	$O(n^{-2})$	$O(n^{-2})$
Framework	Moment-based	Cumulant-based with Bartlett identities
GLM matrix formula	Not available	Equation (60)
Extensions	LR statistic only	LR, score, and Wald statistics
Power preservation	Same local power to $O(n^{-1/2})$	Same local power to $O(n^{-1/2})$

12 References

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